

A certified order-17 exclusion for a C_4 -free common-neighbor problem

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Abstract

For a finite simple graph G , let $c(G)$ count unordered pairs of distinct vertices having exactly one common neighbor. We prove that every C_4 -free graph on exactly 17 vertices with $5 \leq \Delta(G) \leq 9$ satisfies $c(G) \leq 120$. This excludes the order-17 case of the nonplanar alternative in Corollary 5.8 of Ascolese, Negrini, Pagani and Pellegrini. The proof is computer-assisted: elementary pair-packing inequalities reduce the problem to 49 degree profiles, exact rational separating inequalities exclude 46, and checked DRAT refutations exclude the remaining three. A portable supplement contains the certificates and a verifier that reconstructs the finite degree domain, the allocation constraints and every residual CNF. The result concerns this finite order and degree window only.

1 Statement and context

All graphs here are finite, undirected and simple. Write $N_G(x)$ for the neighborhood of x , $d(x) = |N_G(x)|$, and $\Delta(G) = \max_x d(x)$. Define

$$c(G) = \#\{ \{x, y\} \subseteq V(G) : x \neq y, |N_G(x) \cap N_G(y)| = 1 \}.$$

A graph is C_4 -free when it has no four-cycle as a subgraph; the cycle need not be induced. Triangles are allowed throughout.

Theorem 1. *Every C_4 -free graph G on exactly 17 vertices with $5 \leq \Delta(G) \leq 9$ satisfies $c(G) \leq 120$. Consequently no nonplanar graph in this class has $c(G) > \binom{16}{2} = 120$.*

Ascolese, Negrini, Pagani and Pellegrini study reconstruction from common neighbors in [1]. Their Corollary 5.8 reports exhaustive Magma/nauty verification for orders 5 through 16. For larger orders, one remaining alternative for a graph with $c(G) > \binom{|V(G)|-1}{2}$ is nonplanarity together with

$$|V(G)| \geq 17, \quad \lceil \sqrt{|V(G)|} \rceil \leq \Delta(G) \leq \frac{|V(G)| + 1}{2}.$$

They conjecture immediately after that corollary that this alternative never occurs. At order 17 the displayed window is precisely [5, 9]. Theorem 1 answers the corresponding finite existence question negatively, without imposing planarity in the exclusion itself. It verifies this order of the source's nonexistence conjecture; it does not contradict that conjecture.

The common-neighbor identity and pair-packing inequality used below also appear in [1, Lemma 3.1 and Lemma 5.3]. We include their short arguments so the mathematical reduction

does not depend on an external lemma. The contribution beyond the computation reported in that source is a complete, replayable exclusion at order 17, with the degree window specified in Theorem 1. Adjacent examples of finite reductions with checkable certificates are Zeng’s work on regular graphs of prescribed girth [2] and certified SAT encodings for the Erdős–Szekeres problem [3]. These references provide computational-methodology context; neither is used to infer a result about the present graph class.

The literature comparison is limited to the inspected version of [1] and the bounded source search recorded with the research evidence on September 9, 2026. It does not establish exclusive priority. No assertion at orders greater than 17, no equality classification, and no proof-assistant formalization is made here.

2 Reduction to 49 degree profiles

Assume, for a contradiction, that a graph in the theorem’s degree window has $c(G) \geq 121$. If its connected components have orders $n_1, \dots, n_r$ with $r \geq 2$, then

$$c(G) \leq \sum_{i=1}^r \binom{n_i}{2} \leq \binom{16}{2} = 120.$$

The first inequality holds because a pair with a common neighbor belongs to one component. For the second, merging components increases the sum; with two positive parts, moving vertices from the smaller to the larger increases it until the parts are 1 and 16. Thus the hypothetical graph is connected, and all its degrees belong to $\{1, \dots, 9\}$.

A graph contains a four-cycle if and only if some pair of distinct vertices has two distinct common neighbors. Hence all codegrees in a C_4 -free graph are zero or one. Counting paths of length two by their middle vertex gives

$$C := c(G) = \sum_{x \in V(G)} \binom{d(x)}{2}, \quad 121 \leq C \leq \binom{17}{2} = 136. \quad (1)$$

The degree sum $S = \sum_x d(x)$ is even.

Lemma 1. *For every vertex x of a connected C_4 -free graph of order $n \geq 2$,*

$$\sum_{y \neq x} \binom{d(y) - 1}{2} \leq \binom{n - d(x)}{2}.$$

Proof. Put $B = V(G) \setminus N_G(x)$, so $|B| = n - d(x)$; in particular $x \in B$. For $y \neq x$, the set $N_G(y) \cap N_G(x)$ has size at most one, so y has at least $d(y) - 1$ neighbors in B . Pairs of these neighbors cannot be contributed by two distinct vertices y , since such a pair would have two common neighbors. There are only $\binom{|B|}{2}$ available pairs. All degrees are positive, so every binomial coefficient has a nonnegative upper argument. $\square$

Set $D = \Delta(G)$ and $T = \sum_x \binom{d(x)-1}{2}$. Applying Lemma 1 at a vertex of maximum degree yields

$$T \leq \binom{17 - D}{2} + \binom{D - 1}{2}. \quad (2)$$

Let a_j be the number of vertices of degree j . Enumerate all nonnegative integer vectors $(a_1, \dots, a_9)$ with $\sum_j a_j = 17$. There are $\binom{25}{8} = 1,081,575$ such vectors.

Cumulative condition	Histograms retained
$\sum_{j=1}^9 a_j = 17, a_j \geq 0$	1,081,575
$5 \leq D \leq 9, 121 \leq C \leq 136$	47,920
S even	23,946
Maximum-vertex pair packing (2)	49

Table 1: Exact degree-domain enumeration. Every surviving profile has $D = 5$.

For each vector, compute $D = \max\{j : a_j > 0\}$, $S = \sum_j j a_j$, $C = \sum_j a_j \binom{j}{2}$ and $T = \sum_j a_j \binom{j-1}{2}$. Retain it exactly when $5 \leq D \leq 9$, $121 \leq C \leq 136$, S is even, and (2) holds.

The verifier performs this enumeration by choosing the eight bar positions among 25 positions. If $b_1 < \dots < b_8$ are chosen from $\{0, \dots, 24\}$, put $b_0 = -1$, $b_9 = 25$, and $a_j = b_j - b_{j-1} - 1$. This is a bijection with the required histograms. The exact counts are in Table 1. The resulting sorted degree sequences agree with the 49 profiles in the supplement's `evidence/degree-screen.json`. All have maximum degree 5. Indices $0, \dots, 48$ below refer to their lexicographic order as nondecreasing degree sequences. No graphicality criterion is needed for this reduction: these are only necessary conditions, and the enumeration covers the entire hypothetical degree domain.

3 Exact contradictions for neighborhood allocations

Fix a surviving profile and its class sizes $a_1, \dots, a_5$. For a vertex of degree d , define its neighborhood type $h = (h_1, \dots, h_5)$ by letting h_j count its neighbors of degree j . Every realized type belongs to the finite set

$$\mathcal{H}_d = \left\{ h \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 h_j = d, \quad h_j \leq a_j - \mathbf{1}_{j=d}, \quad \sum_{j=1}^5 (j-1)h_j \leq 16 \right\}. \quad (3)$$

The last condition counts nonreturning paths of length two starting at that vertex. Their endpoints are distinct and different from the starting vertex, since a repeated endpoint would give two common neighbors. An endpoint may be adjacent to the start; triangles are not excluded.

Let $x_{d,h}$ count the degree- d vertices having type h . These are nonnegative integers, but we will use only their nonnegativity. The following system is necessary:

$$\sum_{h \in \mathcal{H}_d} x_{d,h} = a_d, \quad 1 \leq d \leq 5, \quad (4)$$

$$\sum_{h \in \mathcal{H}_d} h_e x_{d,h} - \sum_{h \in \mathcal{H}_e} h_d x_{e,h} = 0, \quad 1 \leq d < e \leq 5, \quad (5)$$

$$\sum_{d,h} \binom{h_j}{2} x_{d,h} \leq \binom{a_j}{2}, \quad 1 \leq j \leq 5, \quad (6)$$

$$\sum_{d,h} h_j h_k x_{d,h} \leq a_j a_k, \quad 1 \leq j < k \leq 5. \quad (7)$$

Here $\sum_{d,h}$ ranges over $1 \leq d \leq 5$ and $h \in \mathcal{H}_d$; classes of size zero can be omitted. Equation (5) counts edges between two degree classes in the two directions. Inequalities (6)–(7) count unordered endpoint pairs in prescribed degree classes. Each pair can be joined through at most one middle vertex. They apply whether or not the endpoint pair itself is an edge.

Replace each equality with its two inequalities and write the system as $Ax \leq b$, $x \geq 0$. A separating certificate is a rational vector y indexed by the rows satisfying

$$y \geq 0, \quad yA \geq 0 \quad \text{coordinatewise}, \quad yb = -1. \quad (8)$$

Any nonnegative solution would then satisfy $0 \leq (yA)x \leq yb = -1$, a contradiction.

The file `evidence/allocation-farkas.json` supplies such a vector for each of the 46 indices other than 39, 44 and 48. Every multiplier is encoded as an integer or a rational numerator and denominator. The checker regenerates all types by weak compositions, constructs every row from (4)–(7), and checks every coordinate and right-hand side in (8) using exact rational arithmetic. It also checks that the 46 certified indices and the three residual indices partition all 49 profiles.

For an explicit illustration, profile 0 has $(a_1, a_2, a_3, a_4, a_5) = (4, 1, 0, 0, 12)$. Denote the left-hand sides of (4), (5) and (6) by L_d , B_{de} and P_{jj} , respectively. Its certificate is

$$\frac{3}{25}L_1 + \frac{1}{5}L_2 - \frac{9}{25}L_5 - \frac{3}{25}(B_{12} + B_{15} + B_{25}) + \frac{1}{25}P_{55}.$$

It is a nonnegative combination of the chosen inequality rows, including negative directions for the equalities. Its right-hand side is

$$\frac{3}{25}4 + \frac{1}{5}1 - \frac{9}{25}12 + \frac{1}{25}\binom{12}{2} = -1,$$

and all coefficients on the 14 admissible type variables are nonnegative, as checked exactly. The same test is performed for every stored separator. Floating-point linear optimization was used only to propose multipliers during discovery; no floating-point infeasibility decision is part of these 46 exclusions.

4 The three residual graph formulas

The remaining profiles are in Table 2; exponents denote multiplicity. For each profile, label vertices $0, \dots, 16$ in nondecreasing degree order and introduce one Boolean variable e_{ij} for each $i < j$. Its truth means that $\{i, j\}$ is an edge. Impose these constraints:

1. Exactly d_i of the 16 variables incident with vertex i are true.
2. Each four-element vertex set has three distinct undirected four-cycles. For each cycle, a clause negates its four edge variables. There are $3\binom{17}{4} = 7,140$ such clauses.
3. For each vertex i and each d_i -element subset U of the other vertices with $\sum_{j \in U} (d_j - 1) > 16$, include $\bigvee_{j \in U} \neg e_{\min(i,j), \max(i,j)}$. This follows from the last inequality in (3).
4. Impose the degree-preserving lexicographic comparisons described below.

No connectedness or planarity constraint is imposed on these formulas.

Index	Degree multiset	Edges	C	Variables	Clauses
39	$2^1 4^{10} 5^6$	36	121	3,361	33,907
44	$3^3 4^7 5^7$	36	121	2,911	31,618
48	$4^{11} 5^6$	37	126	3,693	39,135

Table 2: Residual profiles and complete CNF sizes. Each formula has a separately checked DRAT refutation. The count C is forced by the degree profile.

Exact-degree encoding. List incident variables in increasing order of the other endpoint. Let $t_{r,k}$ mean that at least k of the first r variables are true. Use $t_{r,0} = \text{true}$ and $t_{0,k} = \text{false}$ for $k > 0$, computing thresholds through $k = d_i + 1$. For the r th incident variable x , impose

$$s \longleftrightarrow a \vee (b \wedge x), \quad s = t_{r,k}, \quad a = t_{r-1,k}, \quad b = t_{r-1,k-1}.$$

Its four clauses are

$$\begin{aligned} &(\neg a \vee s), & (\neg b \vee \neg x \vee s), \\ &(\neg s \vee a \vee b), & (\neg s \vee a \vee x). \end{aligned}$$

The units t_{16,d_i} and $\neg t_{16,d_i+1}$ force degree exactly d_i . A dedicated variable fixed true supplies the boundary constants. Every graph of the prescribed degrees extends to this encoding by assigning the actual prefix thresholds.

Symmetry and preservation of existence. Order edge coordinates lexicographically by (i, j) and set false $<$ true. For each transposition τ of two vertices in the same degree class, require the edge vector to be no larger than its image under τ . The numbers of comparisons for profiles 39, 44 and 48 are 60, 45 and 70, respectively.

Permutations within the prescribed degree classes preserve all the first three constraints. Each orbit under this group has a lexicographically least edge vector. That vector satisfies every imposed comparison simultaneously. Thus at least one representative of every possible realization remains. Unique representatives are not required. Coordinates fixed by a transposition can be deleted from its comparison because they are equal on both sides.

At a retained coordinate, let x, y be the two compared bits and let p mean that all earlier retained coordinates agree, initially true. The clause $\neg p \vee \neg x \vee y$ forbids the wrong strict order. The next prefix variable q satisfies $q \leftrightarrow p \wedge (x = y)$, encoded by

$$\begin{aligned} &(\neg q \vee p), & (\neg q \vee \neg x \vee y), & (\neg q \vee x \vee \neg y), \\ &(\neg p \vee x \vee y \vee q), & (\neg p \vee \neg x \vee \neg y \vee q). \end{aligned}$$

Assigning prefix agreement its intended meaning extends every lex-admissible edge vector. Consequently any hypothetical graph with a residual profile would give a satisfying assignment to that CNF.

Exact refutations and completion of the proof. CaDiCaL 3.0.1 produced binary DRAT proofs for the three formulas. The separate `drat-trim` program checked each proof against its CNF and returned **VERIFIED** with exit code zero. The verifier independently reconstructs every clause, including auxiliaries, and compares clause multisets with multiplicity against each stored CNF. Its four-cycle construction uses pairs of endpoints and disjoint pairs of common neighbors, rather than

the generator’s three cycles per four-set. Its auxiliary clause templates are derived from complete Boolean truth tables. The reconstruction matches every clause and variable count in Table 2; DRAT replay then establishes unsatisfiability.

Every hypothetical graph is therefore covered by these alternatives: its histogram fails a necessary condition in Section 2; its neighborhood allocation contradicts one of the 46 exact separators in Section 3; or a degree-preserving labeling extends to one of the three refuted formulas in this section. These alternatives exhaust the domain, proving Theorem 1. $\square$

5 Reproducibility and limits

The accompanying `certificates.zip` contains the 49 degree profiles, all rational separators, all three CNF/DRAT pairs, and a standard-library Python verifier. Extract it into a fresh directory and, with `drat-trim` on the executable search path, run

```
python3 verify.py
```

The command checks the supplement’s input hashes, enumerates the full degree domain, checks all 46 rational contradictions, reconstructs the three CNFs, and invokes the proof checker. It writes a machine-readable report and three checker logs under `output/`. Verification needs neither CaDiCaL nor the SciPy and Z3 packages used in discovery. The README specifies the file layout and expected counts.

A separately written verifier was used during mathematical review to reconstruct the reductions and replay all three proofs. The portable verifier retains those reconstruction functions, with a wrapper for the supplement’s file layout and output directory. This separately implemented enumeration and CNF reconstruction uses the same DRAT-checker implementation as the earlier checks. The model-context review is not external peer review.

The computational trust boundary comprises the elementary arguments in this article, the finite enumeration and formula reconstruction, Python’s integer and rational arithmetic, and the DRAT checker’s implementation and runtime. The CNF proofs cover only the three residual profiles; the histogram reduction and rational contradictions are also essential to the full theorem. No solver timeout or unsuccessful search is used as evidence of nonexistence, and no small-order test is extrapolated to order 17.

Computational and AI assistance. OpenAI Codex assisted with mathematical exploration, code generation, certificate analysis, a separate-context mathematical review, and manuscript preparation. Exact arithmetic checks and DRAT proof replay, together with the stated mathematical reductions, supply the computational evidence. Model-generated assessments are not themselves proof certificates.

References

- [1] M. Ascolese, P. Negrini, S. M. C. Pagani and M. A. Pellegrini, *A graph reconstruction problem involving common neighbors*, arXiv:2609.08803v1, 2026. <https://arxiv.org/abs/2609.08803v1>.
- [2] Zijian Zeng, *Proof-Carrying Reductions for 4-Regular Girth-Nine Graphs of Excess Four*, 2026. <https://doi.org/10.2139/ssrn.7374858>.
- [3] Zijian Zeng, *One-Sided Parity Compression for Certified Sat Attacks on the 33-Point Erdős-Szekeres Problem*, 2026. <https://doi.org/10.2139/ssrn.7334758>.